

Nivell 1

Exercici 1

- a. La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit.

```
3  -- NIVEL 1
4  -- Exercici 1
5
6  -- Creamos la tabla credit_card
7  CREATE TABLE credit_card (
8      id VARCHAR(15) PRIMARY KEY,
9      iban VARCHAR(100),
10     pan VARCHAR(50),
11     pin VARCHAR(4),
12     cvv VARCHAR(3),
13     expiring_date VARCHAR(8)
14 );
15
16 ALTER TABLE transaction
17 ADD CONSTRAINT fk_transaction_credit_card
18 FOREIGN KEY (credit_card_id)
19 REFERENCES credit_card(id);
20
21
```

100% 28:19

Action Output

	Time	Action	Response	Duration / Fetch Time
✓	49...	18:53:33 INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS...	1 row(s) affected	0.00011 sec
✓	50...	18:53:33 INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS...	1 row(s) affected	0.00011 sec
✓	50...	18:53:33 INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS...	1 row(s) affected	0.00011 sec
✓	50...	18:53:33 INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS...	1 row(s) affected	0.00011 sec
✗	50...	18:53:48 ALTER TABLE transaction ADD CONSTRAINT fk_transaction_credit_card FOREI...	Error Code: 1452. Cannot add or update a child row: a foreign key constr...	0.286 sec
✓	50...	18:54:14 SELECT t.credit_card_id FROM transaction t LEFT JOIN credit_card c ON t.credi...	1 row(s) returned	0.082 sec / 0.000017...
✓	50...	18:54:57 INSERT INTO credit_card (id) VALUES ('CcU-9999')	1 row(s) affected	0.0015 sec
✓	50...	18:55:02 ALTER TABLE transaction ADD CONSTRAINT fk_transaction_credit_card FOREI...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	3.679 sec

Nota:

He definido *expiring_date* como VARCHAR porque los datos proporcionados en el archivo de carga utilizan el formato (MM/DD/YY) que no es directamente compatible con el tipo DATE de MySQL. Teniendo en cuenta que en este ejercicio el campo se utiliza para almacenar la fecha de vencimiento de la tarjeta y no para realizar operaciones temporales, me resulta más práctico y consistente conservarlo como texto. Al momento de manipular este dato, podría convertirlo en fecha con `STR_TO_DATE('10/30/22', '%m/%d/%y')`.

- b. La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company").

```

1 • ALTER TABLE transaction
2   ADD CONSTRAINT fk_transaction_credit_card
3   FOREIGN KEY (credit_card_id)
4   REFERENCES credit_card(id);

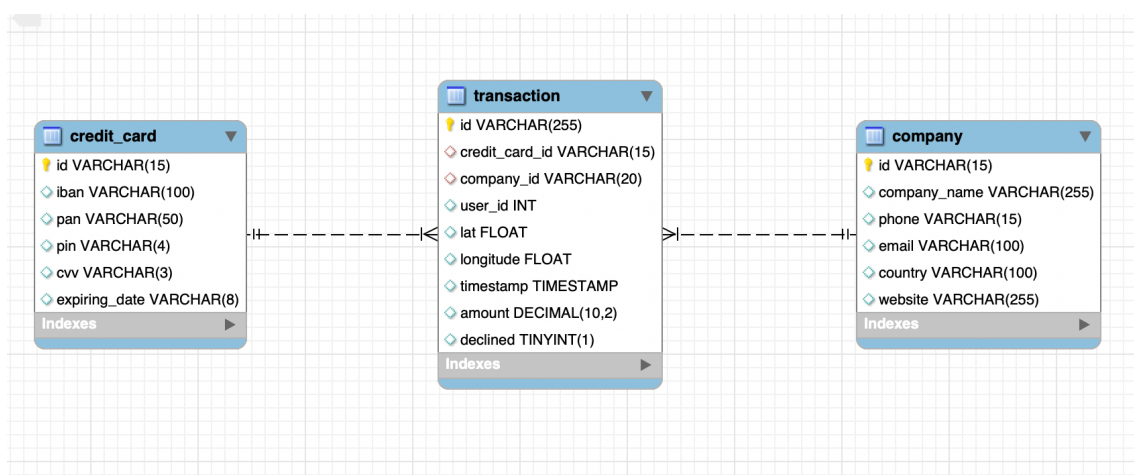
```

Action Output				
	Time	Action	Response	Duration / Fetch Time
✓ 50...	15:37:53	INSERT INTO credit_card (id, iban, pan, pin, cv...	1 row(s) affected	0.00011 sec
✓ 5011	15:37:53	INSERT INTO credit_card (id, iban, pan, pin, cv...	1 row(s) affected	0.00011 sec
✓ 5012	15:37:53	INSERT INTO credit_card (id, iban, pan, pin, cv...	1 row(s) affected	0.000093 sec
✓ 5013	15:37:53	INSERT INTO credit_card (id, iban, pan, pin, cv...	1 row(s) affected	0.000096 sec
✓ 50...	15:37:53	INSERT INTO credit_card (id, iban, pan, pin, cv...	1 row(s) affected	0.00011 sec
✓ 5015	15:40:35	SHOW tables FROM transactions	3 row(s) returned	0.0016 sec / 0.0000
✓ 50...	15:41:02	SHOW columns FROM transaction	9 row(s) returned	0.0027 sec / 0.0000
✓ 5017	15:41:13	SELECT * FROM transaction LIMIT 0, 50000	50000 row(s) returned	0.0015 sec / 0.058 s
✓ 50...	15:42:51	ALTER TABLE transaction ADD CONSTRAINT f...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	2.542 sec

Nota:

Como la tabla transaction ya había sido creada previamente, no era posible modificar su estructura mediante una nueva instrucción CREATE TABLE. Por esta razón, se utiliza ALTER TABLE para añadir la clave foránea y establecer la relación entre la tabla transaction y credit_card, referenciando la clave primaria (id) de esta última.

- c. Després de crear la taula serà necessari que ingressis la informació del document denominat "dades_introduir_credit". Recorda mostrar el diagrama i realitzar una breu descripció d'aquest.



En este diagrama se representa un modelo relacional donde la tabla *transaction* actúa como entidad central (fact table).

Esta tabla contiene como clave primaria el identificador único de cada transacción (id) y dos claves foráneas (credit_card_id y company_id) que establecen la relación con las tablas *credit_card* y *company*, respectivamente.

La relación entre *transaction* y *company* es de uno a muchos (1:N), ya que una empresa puede estar asociada a múltiples transacciones. Del mismo modo, la relación entre *credit_card* y *transaction* también es de uno a muchos (1:N), ya que una tarjeta de crédito puede utilizarse en varias transacciones.

Exercici 2

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999. Recorda mostrar que el canvi es va realitzar.

```
-- Exercici 2
-- El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit
-- amb ID CcU-2938.
-- La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999.
-- Recorda mostrar que el canvi es va realitzar.

17 • SELECT *
18 FROM credit_card
19 WHERE id = 'CcU-2938';

22 • UPDATE credit_card
23 SET iban = 'TR323456312213576817699999'
24 WHERE id = 'CcU-2938';
```

id	iban	pin	cvv	expiring_date
CcU-2938	TR323456312213576817699999	3257	984	10/30/22

	Time	Action	Response	Duration / Fetch Time
1	18:34:06	USE transactions	0 row(s) affected	0.00033 sec
2	18:34:11	UPDATE credit_card SET iban = 'TR323456312213576817699999' WHERE id = 'CcU-2938'	1 row(s) affected Rows matched: 1 Changed: 1 Warni...	0.0019 sec
3	18:34:18	SELECT * FROM credit_card WHERE id = 'CcU-2938' LIMIT 0, 50000	1 row(s) returned	0.00080 sec / 0.000...

Nota: Antes de ingresar el nuevo dato solicitado, se revisa que el registro sea efectivamente incorrecto. Al comprobarlo, se procede a hacer la corrección del mismo.

Exercici 3

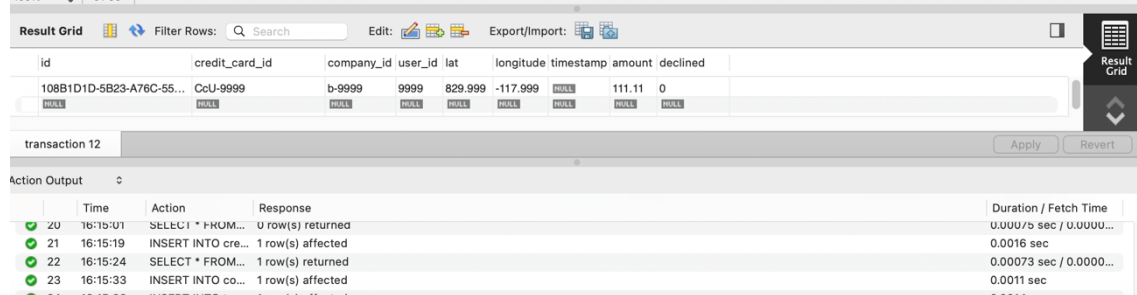
En la taula "transaction" ingressa una nova transacció amb la següent informació:

Id 108B1D1D-5B23-A76C-55EF-C568E49A99DD
credit_card_id CcU-9999
company_id b-9999
user_id 9999
lat 829.999
longitude -117.999
amount 111.11

declined 0

```
INSERT INTO credit_card (id)
VALUES ('CcU-9999'); -- Como no existe, la creo para que no falle la FK

43
44 • SELECT *
45 FROM company
46 WHERE id = 'b-9999';
47
48 • INSERT INTO company (id)
49 VALUES ('b-9999');
50
51 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined)
52 VALUES ('108B1D1D-5B23-A76C-55EF-C568E49A99DD', 'CcU-9999', 'b-9999', 9999, 829.999, -117.999, 111.11, 0);
53
54 • SELECT *
55 FROM transaction
56 WHERE id = '108B1D1D-5B23-A76C-55EF-C568E49A99DD';
57
58
```



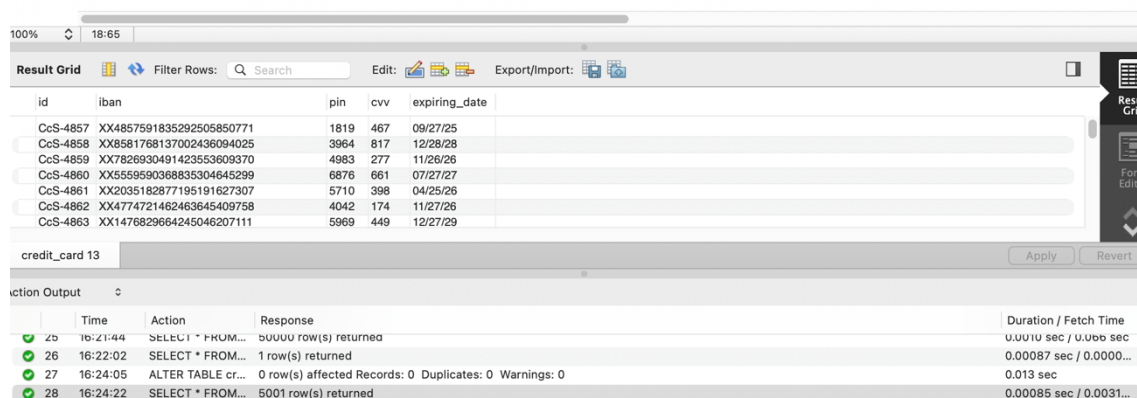
Nota:

El campo timestamp no fue especificado en el enunciado, por lo que se ha dejado con valor NULL.

Exercici 4

Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card. Recordra mostrar el canvi realitzat.

```
58 -- Exercici 4
59 -- Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card. Recordra mostrar el canvi realitzat.
60
61 • ALTER TABLE credit_card
62 DROP pan;
63
64 • SELECT *
65 FROM credit_card;
66
```



Nivell 2

Exercici 1

Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.

```
68 -- NIVEL 2
69 -- Exercici 1
70 -- Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.
71
72 • DELETE FROM transaction
73   WHERE ID = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD';
74
75 • SELECT *
76   FROM transaction
77   WHERE ID = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD';
78
```

Execute the statement under the keyboard cursor

100% 51:77

Result Grid

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

transaction 16

Apply Revert

Action Output

	Time	Action	Response	Duration / Fetch Time
✓ 30	16:30:49	DELETE FROM tr...	1 row(s) affected	0.0026 sec
✗ 31	16:30:54	SELECT * FROM...	Error Code: 1146. Table 'transactions.transactions' doesn't exist	0.00090 sec
✗ 32	16:31:50	SELECT * FROM...	Error Code: 1146. Table 'transactions.transactions' doesn't exist	0.00088 sec
✓ 33	16:32:16	SELECT * FROM...	0 row(s) returned	0.00072 sec / 0.0000...

Exercici 2

La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives. S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves transaccions. Serà necessària que creïs una vista anomenada VistaMarketing que contingui la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia. Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.

```

87 • CREATE VIEW VistaMarketing AS
88     SELECT
89         c.company_name AS nombre_compania,
90         c.phone AS telefono,
91         c.country AS pais,
92         ROUND(AVG(t.amount), 2) AS media_de_compra
93     FROM company AS c
94     JOIN transaction AS t
95     ON c.id = t.company_id
96     GROUP BY c.id;
97
98 • SELECT *
99     FROM VistaMarketing
100     ORDER BY media_de_compra DESC;
101

```

100% 31:100

Result Grid Filter Rows: Search Export:

nombre_compania	telefono	pais	media_de_compra
Ac Fermentum Incorporated	06 85 56 52 33	Germany	284.87
Pretium Neque Corp.	07 77 48 55 28	Australia	276.16
Uma Convallis Associates	06 01 24 77 04	United States	274.24
At Associates	09 56 61 10 65	New Zealand	272.21
Metus Vitae Associates	08 25 44 40 66	Australia	270.08
Aliquet Phas i Limited	09 76 61 47 46	United States	269.69

VistaMarketing 4 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
8	10:30:02	CREATE VIEW InformeTecnicoTransacciones AS SELECT t.id AS ID_transaccion, u.name AS n...	0 row(s) affected	0.0038 sec
9	10:30:08	SELECT * FROM InformeTecnicoTransacciones ORDER BY ID_transaccion DESC LIMIT 0, 50000	50000 row(s) returned	0.0034 sec / 0.245 sec
10	10:33:18	DROP VIEW VistaMarketing	0 row(s) affected	0.0031 sec
11	10:33:42	CREATE VIEW VistaMarketing AS SELECT c.company_name AS nombre_compania, c.pho...	0 row(s) affected	0.0046 sec
12	10:34:05	SELECT * FROM VistaMarketing ORDER BY media_de_compra DESC LIMIT 0, 50000	101 row(s) returned	0.279 sec / 0.000016...

Nota:

Teniendo en cuenta la retroalimentación del sprint 2, agrupo solo por c.id para que la consulta sea más eficiente, ya que, al ser la clave primaria, el resto de los atributos de la tabla dependen funcionalmente de ella.

Exercici 3

Filtra la vista VistaMarketing per a mostrar només les companyies que tenen el seu país de residència en "Germany"

```

102 -- Exercici 3
103 -- Filtra la vista VistaMarketing per a mostrar només les companyies que tenen el seu país de residència en "Germany"
104
105 • SELECT *
106     FROM VistaMarketing
107     WHERE pais = 'Germany';
108

```

100% 24:107

Result Grid Filter Rows: Search Export:

nombre_compania	telefono	pais	media_de_compra
Ac Fermentum Incorporated	06 85 56 52 33	Germany	284.87
Nunc Interdum Incorporated	05 18 15 48 13	Germany	259.32
Convallis In Incorporated	06 66 57 29 50	Germany	257.75
Ac Industries	09 34 65 40 60	Germany	255.15
Rutrum Non Inc.	02 66 31 61 09	Germany	255.14
Auctor Mauris Com.	05 62 87 14 41	Germany	254.77

VistaMarketing 26 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
50	17:12:51	CREATE VIEW VI...	0 row(s) affected	0.0033 sec
51	17:13:46	SELECT c.com...	101 row(s) returned	0.298 sec / 0.000032...
52	17:14:18	SELECT * FROM...	101 row(s) returned	0.236 sec / 0.000065...
53	17:22:24	SELECT * FROM...	8 row(s) returned	0.237 sec / 0.000020...

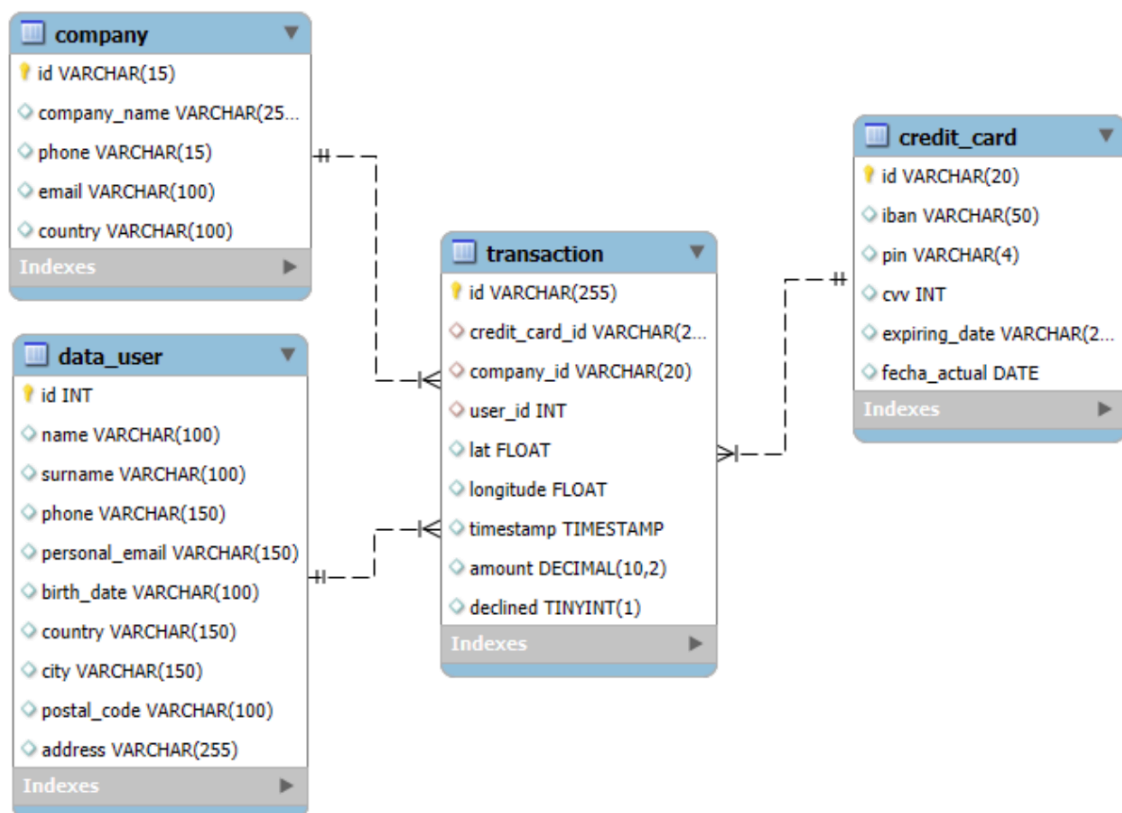
Nota:

Aunque en la tabla original la columna se denomina country, en este caso se utiliza el alias pais, ya que el filtrado se realiza sobre la vista VistaMarketing, donde dicha columna ha sido renombrada.

Nivell 3

Exercici 1

La setmana vinent tindràs una nova reunió amb els gerents de màrqueting. Un company del teu equip va realitzar modificacions en la base de dades, però no recorda com les va realitzar. Et demana que l'ajudis a deixar els comandos executats per a obtenir el següent diagrama:



Recordatori:

En aquesta activitat, és necessari que descriguis el "pas a pas" de les tasques realitzades. És important realitzar descripcions senzilles, simples i fàcils de comprendre. Per a realitzar aquesta activitat hauràs de treballar amb els arxius denominats "estructura_dades_user" i "dades_introduir_user"

Recorda continuar treballant sobre el model i les taules amb les quals ja has treballat fins ara.

PASO A PASO

1. Sobre el archivo en el que se han venido creando las tablas, se crea la tabla de user con la información del archivo “estructura datos user”, para posteriormente establecer su relación con la tabla transaction.

```
1 • - CREATE TABLE IF NOT EXISTS user (  
2     id CHAR(10) PRIMARY KEY,  
3     name VARCHAR(100),  
4     surname VARCHAR(100),  
5     phone VARCHAR(150),  
6     email VARCHAR(150),  
7     birth_date VARCHAR(100),  
8     country VARCHAR(150),  
9     city VARCHAR(150),  
10    postal_code VARCHAR(100),  
11    address VARCHAR(255)  
12 );
```

2. Se revisa que la estructura de la clave primaria (id) de la tabla user sea coherente con la columna user_id de la tabla transaction.
Se observa que en la primera se define como CHAR, mientras que en la segunda se utiliza INT, como se refleja en la siguiente imagen:

```
26 -- Creamos la tabla de user  
27 • - CREATE TABLE IF NOT EXISTS user (  
28     id CHAR(10) PRIMARY KEY,  
29     name VARCHAR(100),  
30     surname VARCHAR(100),  
31     phone VARCHAR(150),  
32     email VARCHAR(150),  
33     birth_date VARCHAR(100),  
34     country VARCHAR(150),  
35     city VARCHAR(150),  
36     postal_code VARCHAR(100),  
37     address VARCHAR(255)  
38 );  
39  
40  
41 -- Creamos la tabla transaction  
42 • - CREATE TABLE IF NOT EXISTS transaction (  
43     id VARCHAR(15) PRIMARY KEY,  
44     credit_card_id VARCHAR(15),  
45     company_id VARCHAR(20),  
46     user_id INT,  
47     lat FLOAT,  
48     longitude FLOAT,  
49     timestamp TIMESTAMP,  
50     amount DECIMAL(10, 2),  
51     declined BOOLEAN,  
52 )
```

Time	Action	Response	Duration / Fetch Time
17:12:51	CREATE VIEW VI...	0 row(s) affected	0.0033 sec
17:13:46	SELECT c.com...	101 row(s) returned	0.298 sec / 0.000032...

Para decidir qué tipo de dato emplear, se revisan los datos del archivo `dades_introduir_user`, comprobando que los valores son numéricos enteros, comenzando desde el registro 151.

Por ello, se decide definir el campo `id` de la tabla `user` como `INT`, manteniendo coherencia con `transaction.user_id` y facilitando la relación entre ambas tablas.

```
19     iban VARCHAR(100),
20     pan  VARCHAR(50),
21     pin  VARCHAR(4),
22     cvv  VARCHAR(3),
23     expiring_date VARCHAR(8)
24 );
25
26 -- Creamos la tabla de user
27 CREATE TABLE IF NOT EXISTS user (
28     id INT PRIMARY KEY,
29     name VARCHAR(100),
30     surname VARCHAR(100),
31     phone VARCHAR(150),
32     email VARCHAR(150),
33     birth_date VARCHAR(100),
34     country VARCHAR(150),
35     city VARCHAR(150),
36     postal_code VARCHAR(100),
37     address VARCHAR(255)
38 );
39
40
```

100% 3:28

Action Output

	Time	Action	Response	Duration / Fetch Time
51	17:13:46	SELECT c.company_name AS nombre_compania, c.phone A...	101 row(s) returned	0.298 sec / 0.000032...
52	17:14:18	SELECT * FROM VistaMarketing LIMIT 0, 50000	101 row(s) returned	0.236 sec / 0.000065...
53	17:22:24	SELECT * FROM VistaMarketing WHERE pais = 'Germany' LIMIT...	8 row(s) returned	0.237 sec / 0.000020...
54	18:03:22	CREATE DATABASE IF NOT EXISTS transactions	1 row(s) affected, 1 warning(s): 1007 Can't create database 'transactions'; database e...	0.00087 sec
55	18:03:22	USE transactions	0 row(s) affected	0.00013 sec
56	18:03:22	CREATE TABLE IF NOT EXISTS company (id VARCHAR(15)...	0 row(s) affected, 1 warning(s): 1050 Table 'company' already exists	0.0027 sec
57	18:03:22	CREATE TABLE IF NOT EXISTS credit_card (id VARCHAR(15)...	0 row(s) affected, 1 warning(s): 1050 Table 'credit_card' already exists	0.00100 sec
58	18:03:22	CREATE TABLE IF NOT EXISTS user (id INT PRIMARY KEY, na...	0 row(s) affected	0.0065 sec

3. Cargamos los datos para la tabla `user` a partir del archivo “datos introducir sprint3 user” al que le añadimos `USE transactions`; para asociarlo a dicha tabla.

Query 1 estructura_datos_credit_card* estructura_dades_credit_card_user* datos introducir sprint3 user*

Limit to 50000 rows

```

1 • USE transactions;
2
3 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "151", "Meghan"
4 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "152", "Hakeem"
5 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "153", "Keegan"
6 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "154", "Cooper"
7 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "155", "Joshua"
8 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "156", "Remedio
9 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "157", "Philip"
10 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "158", "Fatima"
11 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "159", "Kyllynn"
12 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "160", "Lael",
13 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "161", "Nora",
14 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "162", "Frances
15 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "163", "Denton"
16 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "164", "Preston
17 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "165", "Matthew
18 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "166", "Sheila"
19 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "167", "Donna",
20 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "168", "Linda",
21 • INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES ( "169", "Linda",
22 •

```

Action Output

	Time	Action	Response	Duration / Fetch Time
50...	18:08:01	INSERT INTO user (id, name, surname, phone, email, birth_date...	1 row(s) affected	0.00011 sec
50...	18:08:01	INSERT INTO user (id, name, surname, phone, email, birth_date...	1 row(s) affected	0.00011 sec
50...	18:08:01	INSERT INTO user (id, name, surname, phone, email, birth_date...	1 row(s) affected	0.000098 sec
50...	18:08:01	INSERT INTO user (id, name, surname, phone, email, birth_date...	1 row(s) affected	0.000092 sec
50...	18:08:01	INSERT INTO user (id, name, surname, phone, email, birth_date...	1 row(s) affected	0.000096 sec
50...	18:08:01	INSERT INTO user (id, name, surname, phone, email, birth_date...	1 row(s) affected	0.00011 sec
50...	18:08:01	INSERT INTO user (id, name, surname, phone, email, birth_date...	1 row(s) affected	0.000094 sec
50...	18:08:01	INSERT INTO user (id, name, surname, phone, email, birth_date...	1 row(s) affected	0.00010 sec

4. Se verifica que los datos hayan cargado correctamente:

```

110 -- NIVEL 3
111 -- Exercici 1
112 -- La setmana vinent tindrà una nova reunió amb els gerents de màrqueting.
113 -- Un company del teu equip va realitzar modificacions en la base de dades, però no recorda com les va realitzar.
114
115 • SELECT *
116 FROM user;
117
118
119

```

Result Grid

	id	name	surname	phone	email	birth_date	country	city	postal_code	address
1	Zeus	Gamble		1-282-581-0551	interdum.enim@protonmail.edu	Nov 17, 1985	United States	New York	10001	348-7818 Sagittis St.
2	Garrett	Mcconnell		(718) 257-2412	integer.vitae.nibh@protonmail.org	Aug 23, 1992	United States	Philadelphia	19101	903 Sit Ave
3	Ciaran	Harrison		(522) 598-1365	interdum.feugiat@aol.org	Apr 29, 1998	United States	Houston	77001	736-2063 Tellus St.
4	Howard	Stafford		1-411-740-3269	ornare.egestas@icloud.edu	Feb 18, 1989	United States	Phoenix	85001	Ap #545-2244 Erat. Rd.
5	Hayfa	Pierce		1-554-541-2077	et.malesuada.fames@hotmail.org	Sep 26, 1998	United States	Philadelphia	19101	341-2821 Ultrices Av.
6	Joel	Tyson		(718) 288-8020	gravida.nunc.sed@yahoo.ca	Oct 15, 1989	United States	San Jose	95101	888-2799 Amet Street
7	Rafael	Jimenez		(817) 689-0478	egest@outlook.ca	Dec 4, 1981	United States	Chicago	60601	8627 Malesuada Rd.
8	Nissim	Franks		(692) 157-3469	egestas.aliquam.fringilla@google.ca	Aug 1, 1993	United States	New York	10001	Ap #251-7144 Integer St.
9	Mannix	Mcclain		(590) 883-2184	aliquam.nisi@outlook.com	Jan 24, 1987	United States	San Antonio	78201	647-3080 Lacus. St.
10	Robert	Mccarthy		(324) 746-6771	fermentum@protonmail.com	Apr 30, 1984	United States	San Jose	95101	P.O. Box 773

user 27

Action Output

	Time	Action	Response	Duration / Fetch Time
50...	18:08:01	INSERT INTO user (id, name, surname, phone, email, birth_date...	1 row(s) affected	0.00011 sec
50...	18:08:01	INSERT INTO user (id, name, surname, phone, email, birth_date...	1 row(s) affected	0.000094 sec
50...	18:08:01	INSERT INTO user (id, name, surname, phone, email, birth_date...	1 row(s) affected	0.00010 sec
50...	18:10:19	SELECT * FROM user LIMIT 0, 50000	5000 row(s) returned	0.0012 sec / 0.0099...

5. Se añade la restricción de FK a transaction a partir de la siguiente sintaxis:

```

110 -- NIVEL 3
111 -- Exercici 1
112 -- La setmana vinent tindrà una nova reunió amb els gerents de màrqueting.
113 -- Un company del teu equip va realitzar modificacions en la base de dades, però no recorda com les va realitzar.
114
115 • SELECT *
116 FROM user;
117
118 • ALTER TABLE transaction
119 ADD CONSTRAINT fk_transaction_user
120 FOREIGN KEY (user_id)
121 REFERENCES user(id);
122
123

```

	Time	Action	Response	Duration / Fetch Time
✓	50... 18:16:26	SELEC...	50000 row(s) returned	0.091 sec / 0.032 sec
✓	50... 18:17:45	SELEC...	1 row(s) returned	0.00063 sec / 0.000...
✓	50... 18:18:04	SELEC...	1 row(s) returned	0.00049 sec / 0.000...
✓	50... 18:18:43	SELEC...	1 row(s) returned	0.123 sec / 0.000014...
✗	50... 18:21:44	ALTER...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('transactions'.`#sql-1fd_1b`, CONSTRAINT `fk_transaction...` FOREIGN KEY (`user_id`) REFERENCES `user` (`id`))	0.261 sec
✓	50... 18:22:23	SELEC...	50000 row(s) returned	0.091 sec / 0.032 sec
✓	50... 18:22:39	SELEC...	50000 row(s) returned	0.095 sec / 0.032 sec
✓	50... 18:23:26	SELEC...	0 row(s) returned	0.00077 sec / 0.0000...
✗	50... 18:24:02	ALTER...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('transactions'.`#sql-1fd_1b`, CONSTRAINT `fk_transaction...` FOREIGN KEY (`user_id`) REFERENCES `user` (`id`))	0.270 sec

Sin embargo, sale el error:

Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (`transactions`.`#sql-1fd\_1b`, CONSTRAINT `fk\_transaction\_user` FOREIGN KEY (`user\_id`) REFERENCES `user` (`id`)) 0.270 sec

Seguramente el error se debe a que existe un valor en la columna user_id de la tabla transaction que no tiene correspondencia en la tabla user, posiblemente el valor 9999, añadido en el ejercicio 3 del nivel 1.

Para verificar este problema, se realiza un LEFT JOIN con el objetivo de identificar aquellos registros de transaction cuyos user_id no existen en la tabla user.

```

124 • SELECT t.user_id
125 FROM transaction AS t
126 LEFT JOIN user AS u
127 ON t.user_id = u.id
128 WHERE u.id IS NULL;

```

	Time	Action	Response	Duration / Fetch Time
✓	50... 18:18:43	SELEC...	1 row(s) returned	0.123 sec / 0.000014...
✗	50... 18:21:44	ALTER...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('transactions'.`#sql-1fd_1b`, CONSTRAINT `fk_transaction...` FOREIGN KEY (`user_id`) REFERENCES `user` (`id`))	0.261 sec
✓	50... 18:22:23	SELEC...	50000 row(s) returned	0.091 sec / 0.032 sec
✓	50... 18:22:39	SELEC...	50000 row(s) returned	0.095 sec / 0.032 sec
✓	50... 18:23:26	SELEC...	0 row(s) returned	0.00077 sec / 0.0000...
✗	50... 18:24:02	ALTER...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('transactions'.`#sql-1fd_1b`, CONSTRAINT `fk_transaction...` FOREIGN KEY (`user_id`) REFERENCES `user` (`id`))	0.270 sec
✓	50... 18:27:21	SELEC...	1 row(s) returned	0.119 sec / 0.000020...
✗	50... 18:29:38	ALTER...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('transactions'.`#sql-1fd_1b`, CONSTRAINT `fk_transaction...` FOREIGN KEY (`user_id`) REFERENCES `user` (`id`))	0.262 sec
✓	50... 18:29:46	SELEC...	1 row(s) returned	0.116 sec / 0.000009...

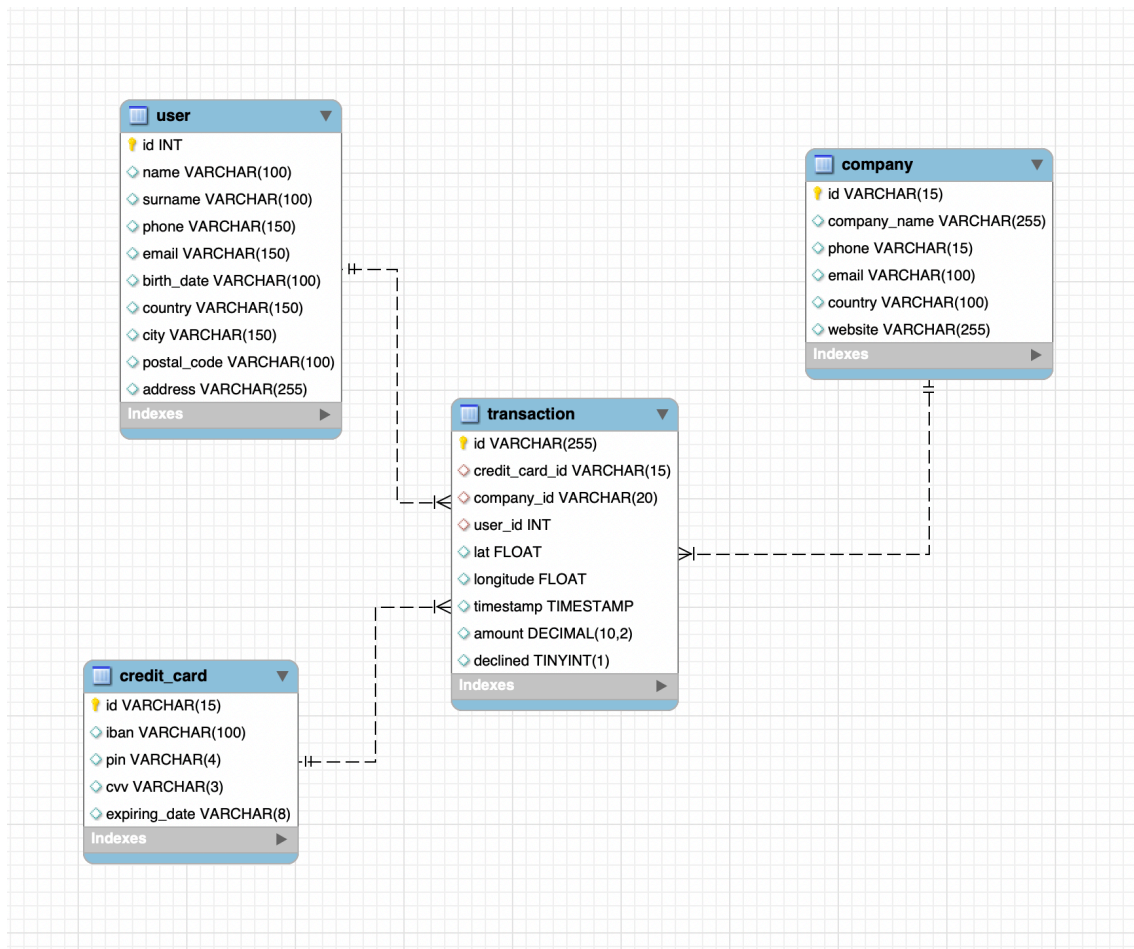
Efectivamente hay un valor nulo que corresponde con la sospecha del ejercicio 3 del Nivel 1.

Para solucionarlo, se crea el usuario con id 9999 en la tabla de user, que tenga todos los demás campos nulos (para llenar después, en caso de que se requiera). Esto permite finalmente ejecutar la restricción de FK de transaction a PK de user.

```
124 • INSERT INTO user (id)
125   VALUES (9999);
126
127 • ALTER TABLE transaction
128   ADD CONSTRAINT fk_transaction_user
129   FOREIGN KEY (user_id)
130   REFERENCES user(id);
```

	Time	Action	Response	Duration / Fetch Time
50...	18:22:23	SELEC...	50000 row(s) returned	0.091 sec / 0.032 sec
50...	18:22:39	SELEC...	50000 row(s) returned	0.095 sec / 0.032 sec
50...	18:23:26	SELEC...	0 row(s) returned	0.00077 sec / 0.0000...
50...	18:24:02	ALTER...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('transactions'. '#sql-1fd_1b', CONSTRAINT 'fk_transaction...	0.270 sec
50...	18:27:21	SELEC...	1 row(s) returned	0.119 sec / 0.000020...
50...	18:29:38	ALTER...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails ('transactions'. '#sql-1fd_1b', CONSTRAINT 'fk_transaction...	0.262 sec
50...	18:29:46	SELEC...	1 row(s) returned	0.116 sec / 0.000009...
50...	18:35:10	INSER...	1 row(s) affected	0.0018 sec
50...	18:35:25	ALTER...	100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0	3.709 sec

- Finalmente se revisa si las tablas quedaron correctamente relacionadas (a través de Database → Reverse Engineer), como se pedía en el diagrama del ejercicio:



Exercici 2

L'empresa també us demana crear una vista anomenada "InformeTecnico" que contingui la següent informació:

ID de la transacció

Nom de l'usuari/ària

Cognom de l'usuari/ària

IBAN de la targeta de crèdit usada.

Nom de la companyia de la transacció realitzada.

Assegureu-vos d'incloure informació rellevant de les taules que coneixereu i utilitzeu àlies per canviar de nom columnes segons calgui.

Mostra els resultats de la vista, ordena els resultats de forma descendent en funció de la variable ID de transacció.

The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, execution, and search, along with a 'Limit to 50000 rows' dropdown. The SQL editor contains the following query:

```
146 SELECT
147     t.id AS ID_transaccion,
148     u.name AS nombre_usuario,
149     u.surname AS apellido_usuario,
150     cc.iban AS IBAN,
151     c.company_name AS nombre_compañia
152 FROM transaction AS t
153 JOIN user AS u
154 ON t.user_id = u.id
155 JOIN credit_card AS cc
156 ON t.credit_card_id = cc.id
157 JOIN company AS c
158 ON t.company_id = c.id;
159
160 • SELECT *
161 FROM InformeTecnico
162 ORDER BY ID_transaccion DESC;
163
```

Below the editor, the 'Result Grid' displays the query results. The first table shows the columns: ID_transaccion, nombre_usuario, apellido_usuario, IBAN, and nombre_compañia. The second table shows the results of the second query, ordered by ID_transaccion in descending order.

ID_transaccion	nombre_usuario	apellido_usuario	IBAN	nombre_compañia
FFFD31D6-9495-47CE-B54A-7DB8E1CC274B	Bmrgli	Tprvmirc	XX794814451211289182490922	Turpis Company
FFFCF76D-ECF0-4985-A2D0-B2A7B75998FC	Dfrled	Vilqjdl	XX636251701647892036676034	Amet Nulla Donec Corporation
FFFC9E8D-27C7-4ADE-98F2-7533EF4DF126	Securp	Faolfvly	XX162677143304223631437567	Nunc Interdum Incorporated
FFFB270D-F53A-4D5D-9666-E5307C53CC84	Ggzjpa	Uirzjuh	XX395114267082019952567052	Viverra Donec Foundation
FFF9E3CE-234E-408C-A8EF-F9CAD577224A	Yshimq	Zpsjsleed	XX8845462156537570367941	Convallis In Incorporated
FFF9F17B-R779-4DF0-66R0-49AF088R0R1	lounny	Ywczwmm	XX321405515711854384711481	Mis Aenean Fret Foundation

The 'Action Output' section at the bottom shows the execution details of the queries:

	Time	Action	Response	Duration / Fetch Time
6	10:26:52	SELECT * FROM InformeTecnico ORDER BY ID_transaccion LIMIT 0, 50000	50000 row(s) returned	0.022 sec / 0.229 sec
7	10:28:05	SELECT * FROM InformeTecnico ORDER BY ID_transaccion DESC LIMIT 0, 50000	50000 row(s) returned	0.0045 sec / 0.236 sec

Nota:

Se ha creado la vista *InformeTecnico* con los datos solicitados en el enunciado. Adicionalmente, se ha creado una segunda versión de informe llamada *InformeTecnicoTransacciones* que se observa en la siguiente imagen.

```

165 • CREATE VIEW InformeTecnicoTransacciones AS
166     SELECT
167         t.id AS ID_transaccion,
168         u.name AS nombre_usuario,
169         u.surname AS apellido_usuario,
170         cc.iban AS IBAN,
171         c.company_name AS nombre_compañía,
172         IF(u.country = c.country, 'Nacional', 'Internacional') AS tipo_transaccion
173     FROM transaction AS t
174     JOIN user AS u
175     ON t.user_id = u.id
176     JOIN credit_card AS cc
177     ON t.credit_card_id = cc.id
178     JOIN company AS c
179     ON t.company_id = c.id;
180
181 • SELECT *
182     FROM InformeTecnicoTransacciones
183     ORDER BY ID_transaccion DESC;
184
100% 30:183

```

ID_transaccion	nombre_usuario	apellido_usuario	IBAN	nombre_compañía	tipo_transaccion
FFFD31D6-9495-47CE-B54A-7DB8E1CC274B	Bmrgli	Tprvmrc	XX794814451211289182490922	Turpis Company	Internacional
FFFO77D0-ECF0-4985-A2D0-B2A7B75998FC	Drifed	Vlqgdl	XX636251701647892036676034	Amet Nulla Donec Corporation	Internacional
FFFO9E8D-27C7-4ADE-98F2-7533EF4DF126	Securp	Facvqfy	XX162677143304223631437567	Nunc Interdum Incorporated	Internacional
FFFB270D-F53A-4D5D-9686-E5307C53CC84	Gqzjpa	Uirzjuhl	XX395114267082019952567052	Viverra Donec Foundation	Internacional
FF9F93CF-234F-40RC-ARFF-F9CA0577224A	Yshimn	Znsidled	XX8845462156537570367941	Convallis In Incorporated	Nacional

InformeTecnicoTransacciones 3

Action Output

	Time	Action	Response	Duration / Fetch Time
9	10:30:08	SELECT * FROM InformeTecnicoTransacciones ORDER BY ID_transaccion DESC LIMIT 0, 50000	50000 row(s) returned	0.0034 sec / 0.245 sec

Esta vista incluye un nuevo campo (tipo_transaccion), que permite clasificar las transacciones como nacionales (cuando usuario y compañía pertenecen al mismo país) o internacionales (cuando pertenecen a países distintos).

Esta clasificación podría aportar valor para futuros análisis de carácter geográfico o de comportamiento de las transacciones a nivel nacional e internacional.

Para la implementación se ha utilizado la función IF, al tratarse de una condición simple y mejorar la legibilidad de la consulta, aunque se es consciente de que CASE es la alternativa estándar en SQL.